In-Class Quiz 1

Course: Classical Mechanics
Duration: 10 minutes
Total Points: 10

Name	Student ID

Question 1

The **Euler–Lagrange equation** for a Lagrangian $L(q, \dot{q}, t)$ is

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0. \tag{1}$$

Explain the Beltrami identity and prove it.

Answer Space